

Numerical Stability of Functional MRI Connectivity Biomarkers

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Project website

This thesis investigates the numerical stability of fMRI connectivity biomarkers. Using floating-point perturbations, it assesses graph metrics, connectivity matrices, and PCA-based feature extraction, demonstrating that key biomarkers for depression research remain robust despite computational variability.



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Context

Functional magnetic resonance imaging (fMRI) enables non-invasive observation of brain activity, making it a cornerstone of modern neuroscience. A key application is the analysis of functional connectivity (FC), which examines temporal correlations between brain regions to construct connectivity matrices. These matrices serve as the foundation for deriving biomarkers used in clinical research, particularly for conditions such as major depressive disorder (MDD).

Small numerical perturbations introduced by factors such as operating system differences, hardware architecture, and parallelization strategies can propagate through the pipeline, potentially affecting downstream results. In clinical applications, where reproducibility is essential, understanding and quantifying this numerical variability is critical.

Approach

This thesis investigates the numerical stability of fMRI connectivity biomarkers at three levels using Monte Carlo Arithmetic (MCA) to introduce controlled floating-point perturbations. A custom Docker container ran the fMRIPrep preprocessing pipeline with MCA instrumentation.

First, the stability of graph-theoretical metrics was assessed by reproducing Alizadeh et al., 2026¹'s work on a different multi-site dataset with a more extensive confound regression strategy (15 vs. 6 confounds), using perturbed fMRIPrep outputs. FC matrices were thresholded at multiple values (0.05 to 0.5) before computing graph metrics.

Second, the analysis was extended to individual edges of the FC matrices to assess edge-wise numerical variability and compare the impact of different confound strategies, particularly global signal regression.

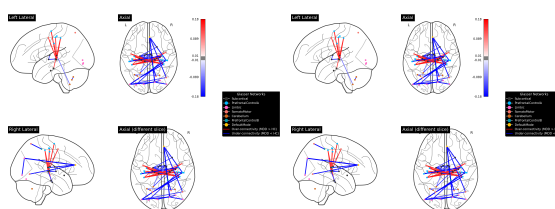
Finally, the stability of Yamashita et al., 2026's² PCA-based feature extraction method for MDD biomarkers was evaluated. The method was applied to the complete SRPB and BMB datasets to assess cross-dataset stability, and then tested under targeted perturbations of FC matrix computation and PCA dimensionality reduction, with inputs cast from 64-bit to 32-bit precision.

Results

Graph metrics reproduction showed consistent trends with the original study across thresholds, but with Numerical-Population Variability Ratio (NPVR) values exceeding the 0.1–0.2 range originally reported. The more extensive confound regression (15 vs. 6 confounds) amplified this variability substantially.

At the edge level, FC matrices exhibited substantial variability, with a mean NPVR of approximately 11% using 15 confounds. Notably, the inclusion of global signal regression halved this variability across most edges, suggesting it acts as a strong stabilizer against floating-point perturbations.

PCA-based feature extraction proved robust across datasets and perturbations. The pattern of MDD underconnectivity remained consistent (prefrontal, motor, and subcortical regions), and selected biomarkers were identical across all 100 perturbed runs on SRPB and 15 on BMB.



SRPB dataset PCA-selected biomarkers: original (left) versus consensus after 100 perturbed runs (right)

Conclusion

Graph metrics were successfully reproduced with consistent trends, though at higher variability than previously reported. Edge-level analysis revealed substantial floating-point sensitivity, mitigated by global signal regression. PCA-based feature extraction proved remarkably robust. Across both datasets, biomarkers showed the same under-connectivity patterns and involved the same brain areas, and under numerical perturbations, the selected biomarkers remained perfectly stable within each dataset. Future work could examine all individual confound effects or apply additional perturbations to the PCA features extraction pipeline.

¹M. Alizadeh, Y. Chatelain, G. Kiar, and T. Glatard, "Numerical Variability of functional MRI Graph Measures," bioRxiv, 2026, doi: [10.64898/2025.12.22.695524](https://doi.org/10.64898/2025.12.22.695524).

²A. Yamashita et al., "Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection," Imaging Neuroscience, vol. 4, p. IMAG.a.1121, 2026, doi: [10.1162/IMAG.a.1121](https://doi.org/10.1162/IMAG.a.1121).